

Summer Term 2026 – Julián Méndez & Ricardo Langner

Interactive Information Visualization

Exercise: **Assignment 1**

Assignment 1

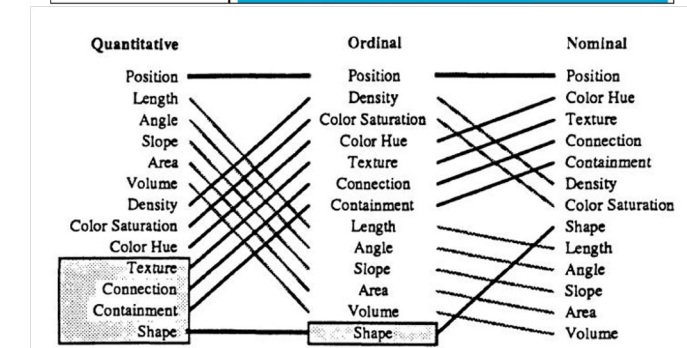
The overall goal of this assignment is to ...

- **Revisit** and **strengthen** knowledge of: How to map abstract and tabular data to visual representations and structures
- Look again into the first three chapters of the DataVis class, find them in our OPAL course
(OPAL > Lecture > Additional Resources > Part of DataVis Lecture)
 - Introduction
 - Perception and Cognition
 - **Mapping to Visual Variables**
- You should have a very good/profound knowledge about the **use of visual variables**

Image source:

https://infovis-wiki.net/wiki/Visual_Variables

Bertin's Original Visual Variables	
Position changes in the x, y location	
Size change in length, area or repetition	
Shape infinite number of shapes	
Value changes from light to dark	
Colour changes in hue at a given value	
Orientation changes in alignment	
Texture variation in 'grain'	



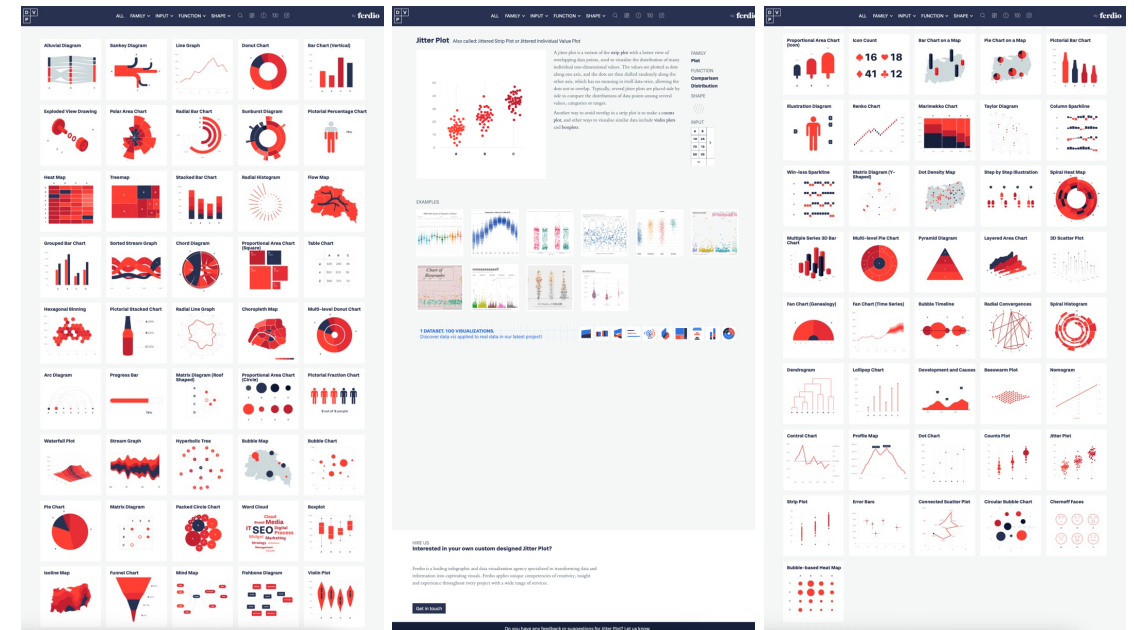
	Size	Shape	Brightness	Color	Orientation	Spacing	Perspective height	Arrangement
Point								
Linear								
Areal								

Assignment 1

Your task

- Draw/sketch visualization applications
- Use only **pen and paper**
- Produce rich ideas
 - Focus on the data visualization, but also describe UI, workflows, interactions, users, usage situations, ...
 - Try out different data mappings and especially different visual variables, don't be afraid of new or crazy concepts

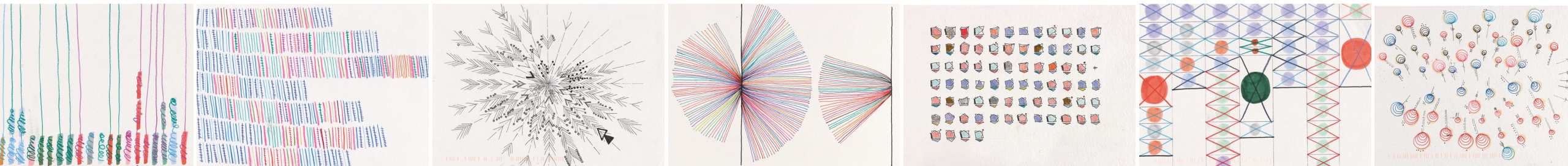
Data Viz Project – Library of data visualizations
screenshots from <https://datavizproject.com/>



Dear Data Project

The power of sketches/drawings

image source <https://www.dear-data.com/theproject>



Assignment 1

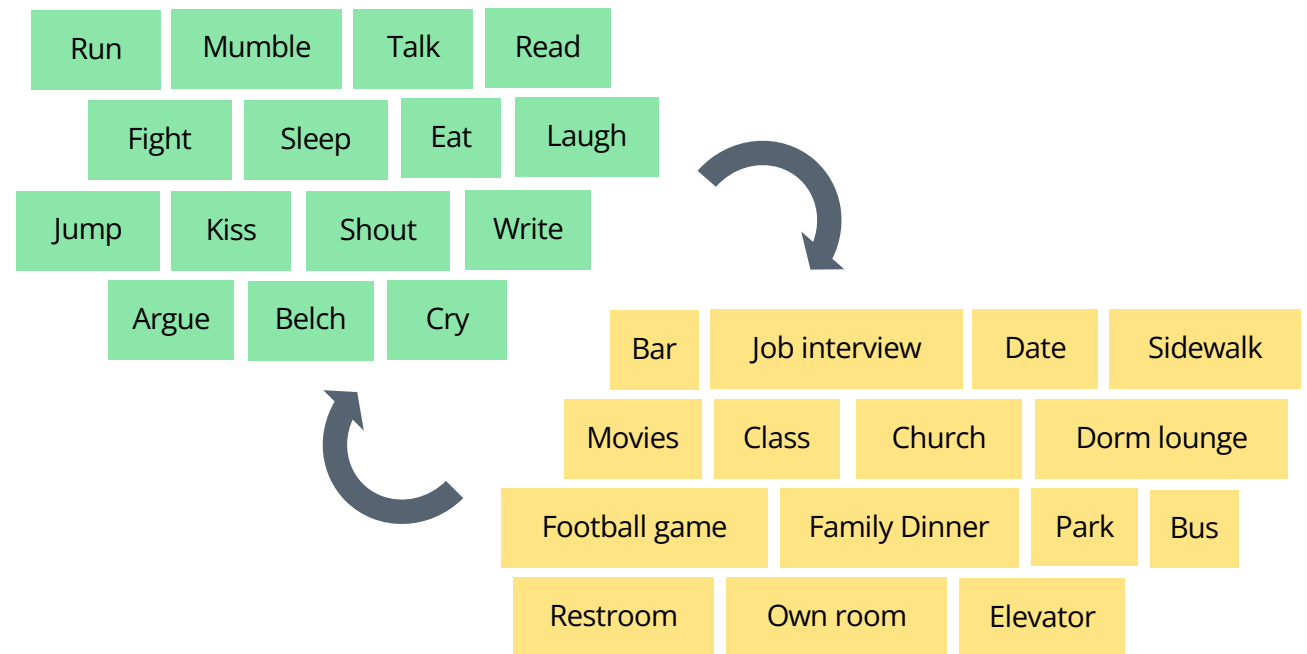
What to visualize? You have two options ...

You find the data at: <https://datashare.tu-dresden.de/s/mJ2KTnieR5GPGds>

Option 1: Soccer players



Option 2: Behavior/Situation combinations



Assignment 1

What to visualize? You have two options ...

You find the data at: <https://datashare.tu-dresden.de/s/mJ2KTnieR5GPGds>

Option 1: **Soccer players**

- Data 2017/18 Champions League season
- **95** soccer players (**nodes**), each up to **40 data attributes**
- **~1050** relations/**edges** between them
- Edge indicate if and how often players have played for the same club during their career (edge weight, val)
- Own interactive visualization: <https://vca.informatik.uni-rostock.de/~ct/software/RMC/>
- Data is not in tabular format (convert)

Option 2: **Behavior/Situation combinations**

- **15 behaviors** and **15 situations**/locations
- Mean appropriateness **ratings of all the 225** different **combinations**
- Values
 - 0 = "The behavior is extremely inappropriate in this situation"
 - 9 = "The behavior is extremely appropriate in this situation"
- No own interactive example visualization, yet 😊
- Data in tabular format

Assignment 1

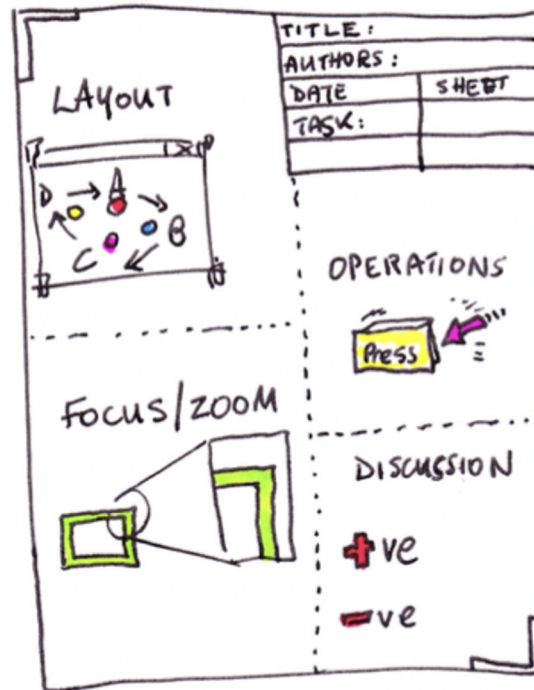
Five Design Sheet Approach (recommended)

Study the approach: <https://fds-design.github.io/>

SHEET 1



SHEET 2,3,4



SHEET 5

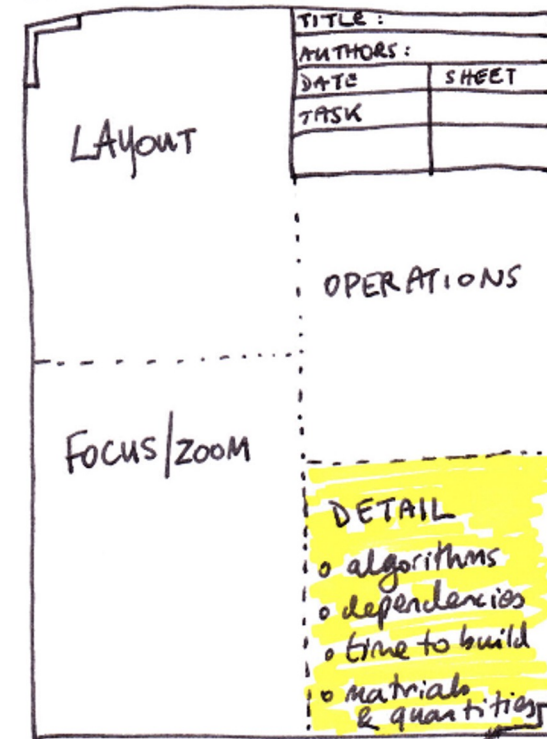


Image source:
<https://fds-design.github.io/>

Assignment 1

Five Design Sheet Approach (recommended)

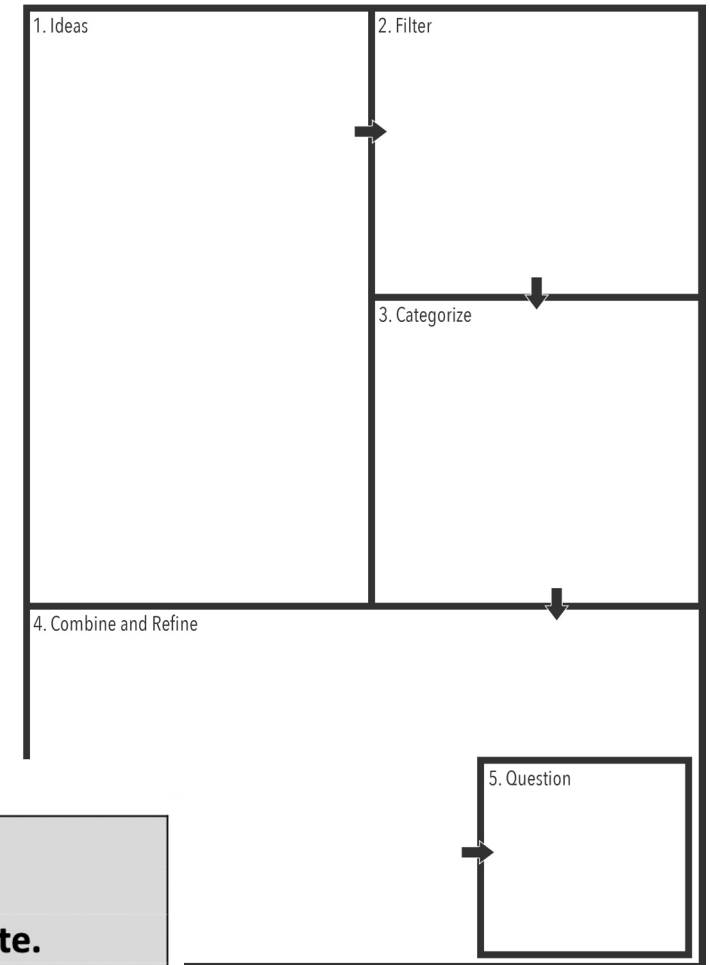
Image source:
<https://fds-design.github.io/>

Phase 1 (Sheet 1): Brainstorming

1. *Ideas*
 - Create, sketch, and generate as many mini-ideas you can
 - You can also use sticky notes
2. *Filter*
 - Get rid of duplicates and very similar ideas
 - Can also be skipped
3. *Categorize*
 - Group or order ideas, based on similarity, functioning, ...
4. *Combine/Refine*
 - Make your mini-ideas “a bit bigger”
 - Add more details, consider special cases, maybe combine them, multiple views
5. *Question*
 - What has been generated?
Maybe think of it as a title

Rules of Brainstorming

1. **Don't criticize**
2. **Don't evaluate.**
3. **Generate-generate-generate.**
4. **Ideate the whole design space.**



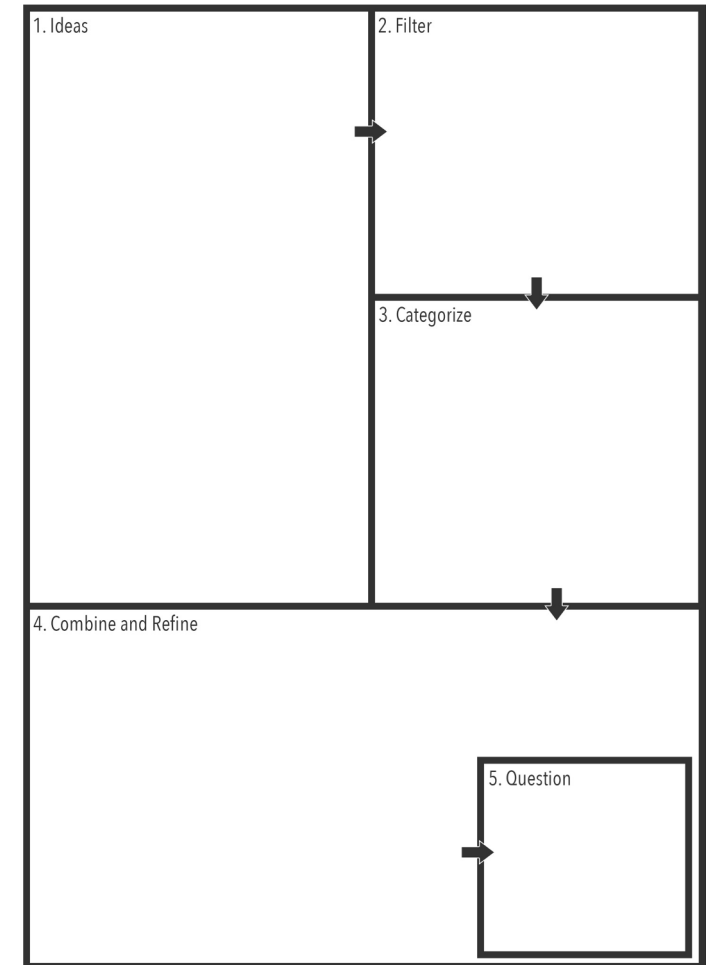
Assignment 1

Five Design Sheet Approach (recommended)

Phase 2 (Sheet 2/3/4): Initial Designs

1. *Ideas* • Application layout, UI, your Vis on a screen
2. *Focus* • App details, highlight/zoom into key elements
3. *Operations* • Describe/sketch how to operate/interact your vis
4. *Discussion* • Describe/sketch pros/cons of your vis or layout
5. *Metadata* • Title, authors, date, sheet number, task (what is it good for?)

Image source:
<https://fds-design.github.io/>



Assignment 1

Five Design Sheet Approach (recommended)

Phase 3 (Sheet 5): Realization Design

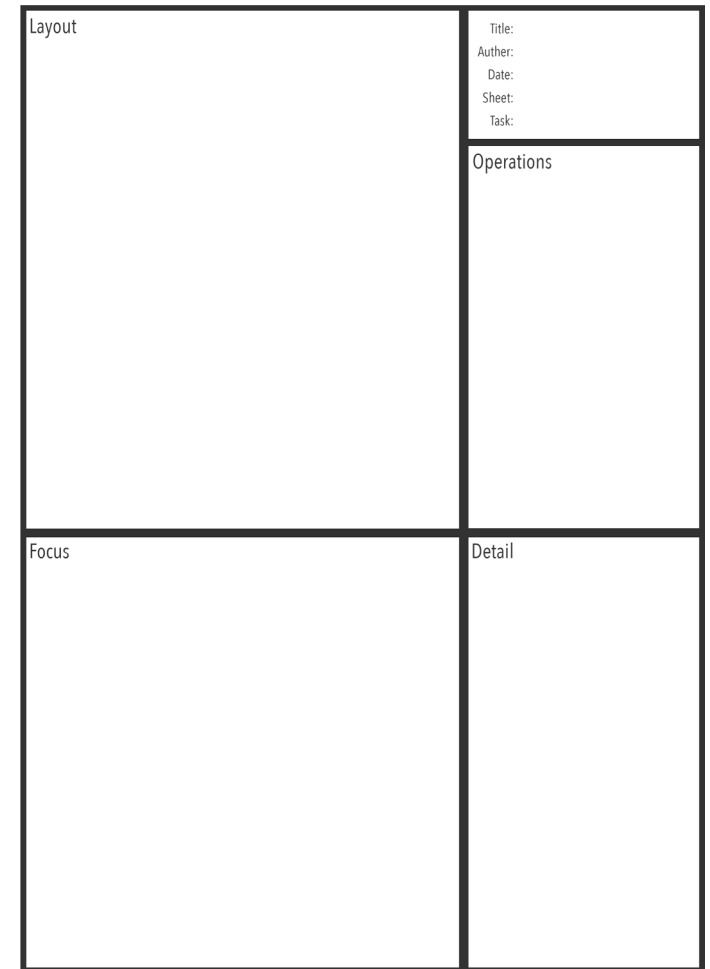
Finalize your design

Basically, only adds details to sheet 2/3/4

- Algorithms, References to useful resources
- Dependencies: what do you need for it?
- Estimated costs/effort
- Other specific prerequisites, requirements, ...

THEN you can usually start developing,
which of course is not part of this assignment 😊

Image source:
<https://fds-design.github.io/>



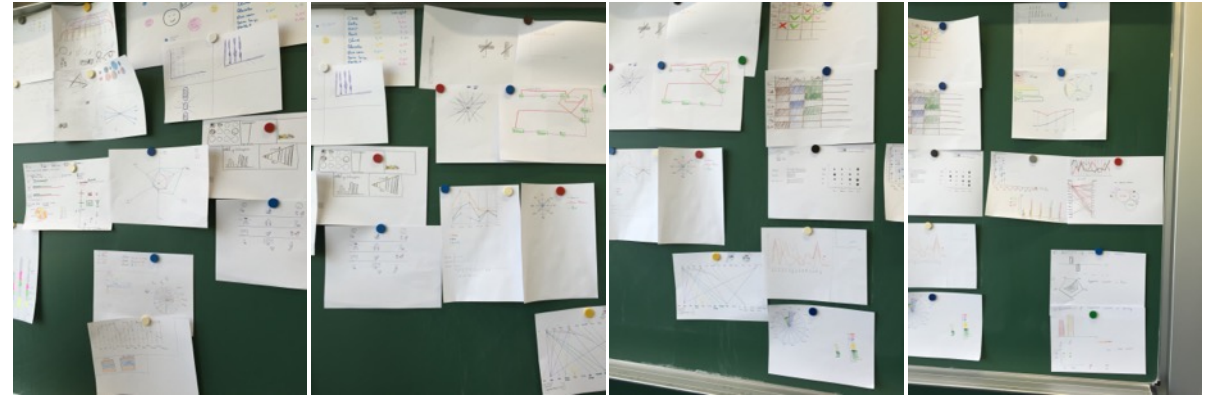
Assignment 1

Summary

Revisit, remember, study the essentials about mapping data to visual variables

Individually and in your team ...

1. Draw/sketch ideas for a data visualization application/tool for one of the data sets
2. Variety: create several variants or alternatives, e.g., by considering different
 - Visualization techniques
 - Visual variables
 - (other factors) input/output devices, users, usage situations, ...
3. Ideally, you follow the FDS approach (or let it inspire your process)
4. Take pictures of the drawings/sketches
5. Share and discuss the designs with your team
6. Upload the archive to the submission folder
7. **Bring your designs with you** (the originals or pictures) to the next on-site exercise, because **we will discuss/reflect about them in group work**



Exercise Interactive Information Visualization

Thank you!

Any questions?

Contact: julian.mendez2@tu-dresden.de
ricardo.langner@tu-dresden.de